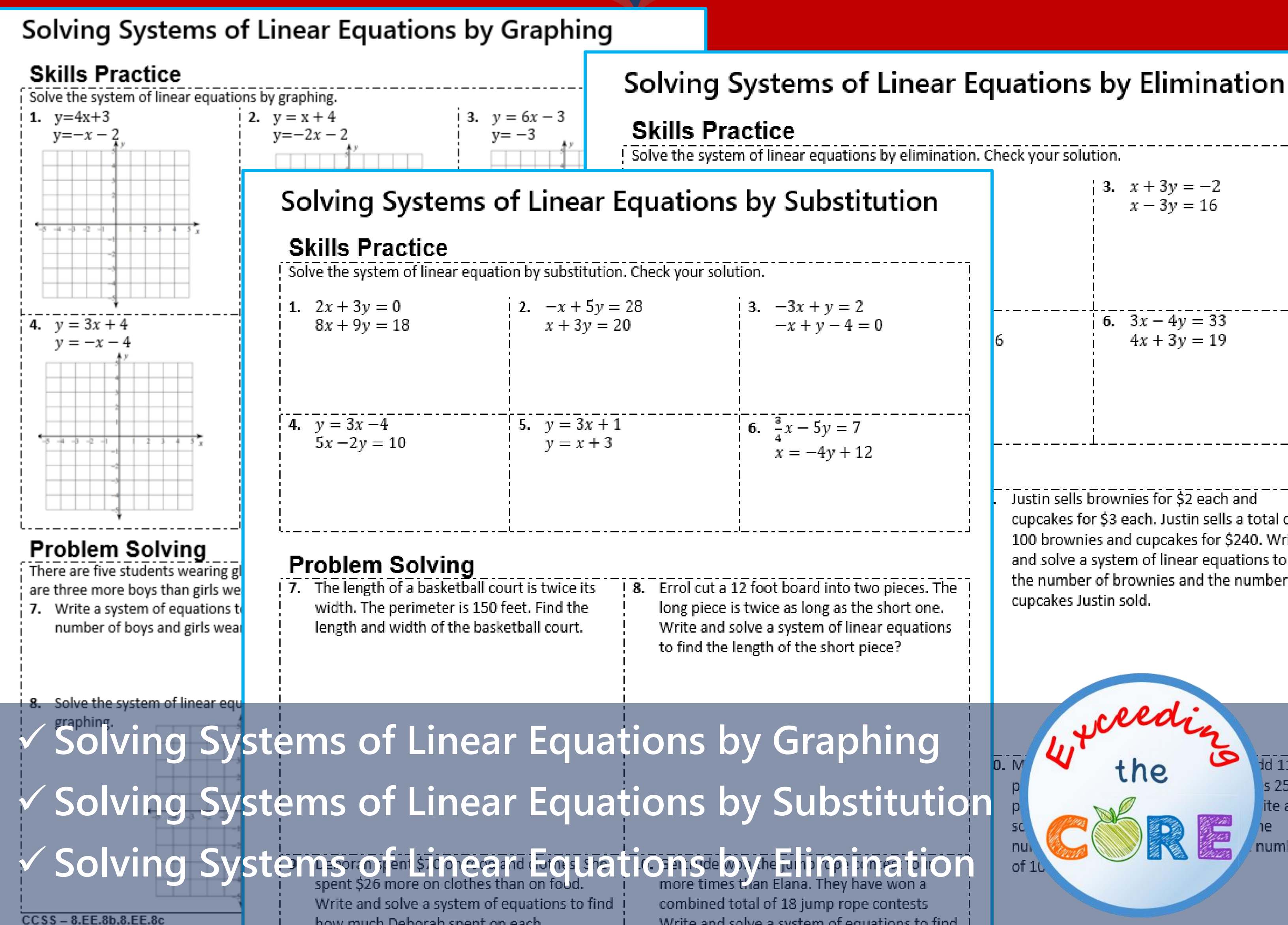
SYSTEMS of LINEAR EQUATIONS

SKILLS PRACTICE & WORD PROBLEMS

PRACTICE WORKSHEETS


Solving Systems of Linear Equations by Graphing

Skills Practice

Solve the system of linear equations by graphing.

1. y=4x+3
y=−x − 2

2. y = x + 4
y=−2x − 2

3. $y = 6x - 3$
$y = -3$

4. $y = 3x + 4$
$y = -x - 4$

Problem Solving

There are five students wearing gl... are three more boys than girls we...

7. Write a system of equations t... number of boys and girls wear...

8. Solve the system of linear equ... graphing.

CCSS – 8.EE.8b,8.EE.8c

Solving Systems of Linear Equations by Elimination

Skills Practice

Solve the system of linear equations by elimination. Check your solution.

3. $x + 3y = -2$
$x - 3y = 16$

6. $3x - 4y = 33$
$4x + 3y = 19$

Justin sells brownies for \$2 each and cupcakes for \$3 each. Justin sells a total o... 100 brownies and cupcakes for \$240. Wri... and solve a system of linear equations to... the number of brownies and the number... cupcakes Justin sold.

Solving Systems of Linear Equations by Substitution

Skills Practice

Solve the system of linear equation by substitution. Check your solution.

1. $2x + 3y = 0$
$8x + 9y = 18$

2. $-x + 5y = 28$
$x + 3y = 20$

3. $-3x + y = 2$
$-x + y - 4 = 0$

4. $y = 3x - 4$
$5x - 2y = 10$

5. $y = 3x + 1$
$y = x + 3$

6. $\frac{3}{4}x - 5y = 7$
$x = -4y + 12$

Problem Solving

7. The length of a basketball court is twice its width. The perimeter is 150 feet. Find the length and width of the basketball court.

8. Errol cut a 12 foot board into two pieces. The long piece is twice as long as the short one. Write and solve a system of linear equations to find the length of the short piece?

... spent \$26 more on clothes than on food. Write and solve a system of equations to find how much Deborah spent on each.

... more times than Elana. They have won a combined total of 18 jump rope contests. Write and solve a system of equations to find...

✓ Solving Systems of Linear Equations by Graphing

✓ Solving Systems of Linear Equations by Substitution

✓ Solving Systems of Linear Equations by Elimination

Exceeding the CORE

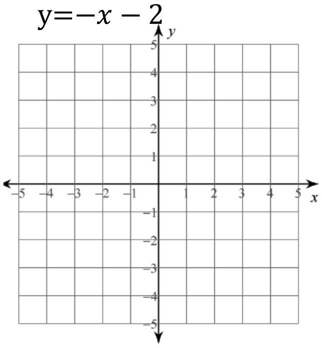
Name ________________________________ Class_________ Date________________________

Solving Systems of Linear Equations by Graphing

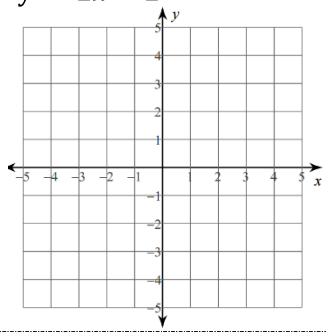
Skills Practice

Solve the system of linear equations by graphing.

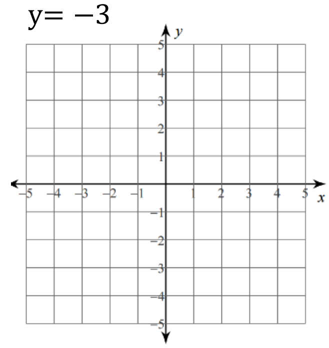
1. y=4x+3
y=$-x-2$



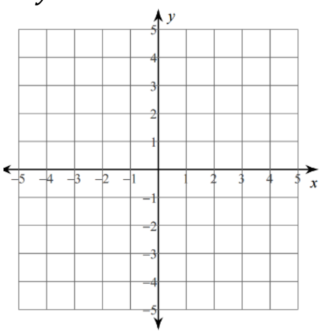
2. y = x + 4
y=$-2x-2$



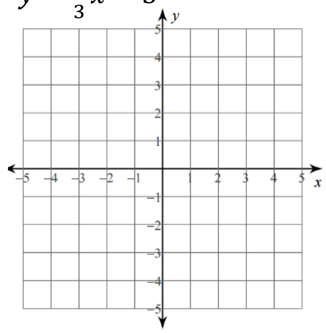
3. $y = 6x - 3$
y= -3



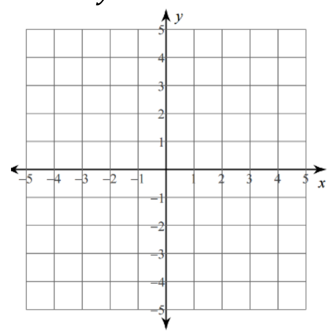
4. $y = 3x + 4$
$y = -x - 4$



5. $y = -x + 1$
$y = \frac{1}{3}x - 3$



6. $-3x + y = 1$
$2x + y = -4$

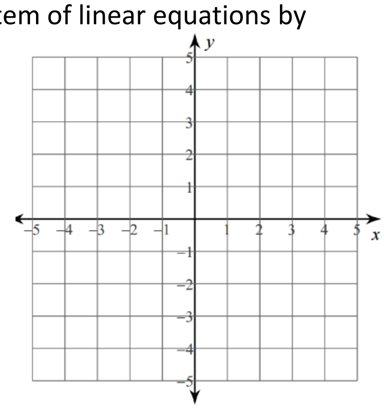


Problem Solving

There are five students wearing glasses. There are three more boys than girls wearing glasses.

7. Write a system of equations to find the number of boys and girls wearing glasses.

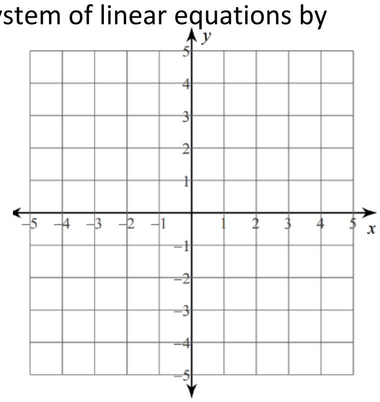
8. Solve the system of linear equations by graphing.



Laura bought a notebook and pencil for $4.50. The notebook cost twice the amount of the pencil.

9. Write a system of equations to find the cost of the notebook and pencil.

10. Solve the system of linear equations by graphing.



CCSS – 8.EE.8a,8.EE.8c

Name ________________________________ Class_________ Date________________________

Solving Systems of Linear Equations by Substitution

Skills Practice

Solve the system of linear equation by substitution. Check your solution.

1. $2x + 3y = 0$ $8x + 9y = 18$	**2.** $-x + 5y = 28$ $x + 3y = 20$	**3.** $-3x + y = 2$ $-x + y - 4 = 0$
4. $y = 3x - 4$ $5x - 2y = 10$	**5.** $y = 3x + 1$ $y = x + 3$	**6.** $\frac{3}{4}x - 5y = 7$ $x = -4y + 12$

Problem Solving

7. The length of a basketball court is twice its width. The perimeter is 150 feet. Find the length and width of the basketball court.

8. Errol cut a 12 foot board into two pieces. The long piece is twice as long as the short one. Write and solve a system of linear equations to find the length of the short piece?

9. Deborah spent \$70 on food and clothes. She spent \$26 more on clothes than on food. Write and solve a system of equations to find how much Deborah spent on each.

10. Gertrude won the jump rope contest four more times than Elana. They have won a combined total of 18 jump rope contests Write and solve a system of equations to find the number of contests Gertrude won and the number of contests Elana won.

CCSS – 8.EE.8b,8.EE.8c

Name ______________________________ Class_________ Date________________________

Solving Systems of Linear Equations by Elimination

Skills Practice

Solve the system of linear equations by elimination. Check your solution.

1. $x + 2y = 4$
$-x - y = 2$

2. $2x + y = 3$
$x - y = 3$

3. $x + 3y = -2$
$x - 3y = 16$

4. $-5x + 2y = 13$
$5x + y = -1$

5. $3x - y = 14$
$-3x + 4y = 16$

6. $3x - 4y = 33$
$4x + 3y = 19$

Problem Solving

7. You buy 5 pairs of jeans and 10 t-shirts for \$140. Your friend buys 3 pairs of jeans and 5 t-shirts for \$79. Write and solve a system of linear equations to find the cost of each t-shirt.

8. Justin sells brownies for \$2 each and cupcakes for \$3 each. Justin sells a total of 100 brownies and cupcakes for \$240. Write and solve a system of linear equations to find the number of brownies and the number of cupcakes Justin sold.

9. The seventh and eighth grade students collected canned goods to donate to a local charity. Together they collected 204 canned goods. The eighth grade collected 3 times as many canned goods as the seventh grade. Write and solve a system of linear equations to find how many canned goods the eighth grade students collected.

10. Marcelle uses a total of 6 plates to add 110 pounds to a weight lifting bar. He uses 25-pound plates and 5-pound plates. Write and solve a system of equations to find the number of 25-pound plates and the number of 5-pound plates he uses.

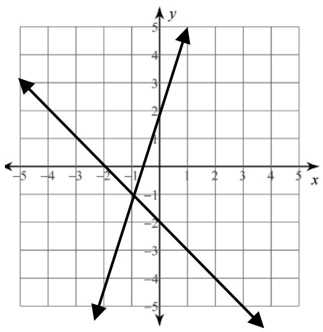
CCSS – 8.EE.8b,8.EE.8c

Name ______________________________ Class__________ Date__________________________

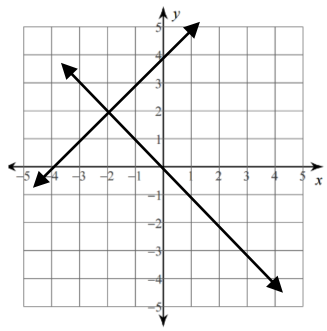
Systems of Linear Equations Practice Worksheets - ANSWER KEY

Solving Systems of Linear Equations by Graphing

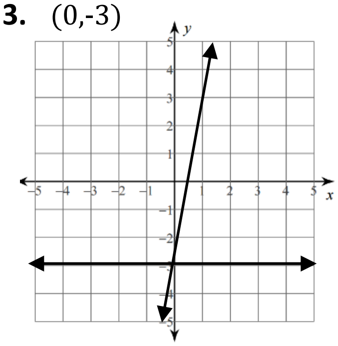
1. (-1,-1)



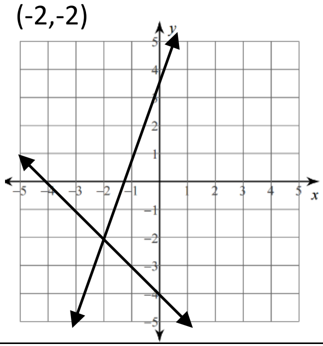
2. (-2,2)



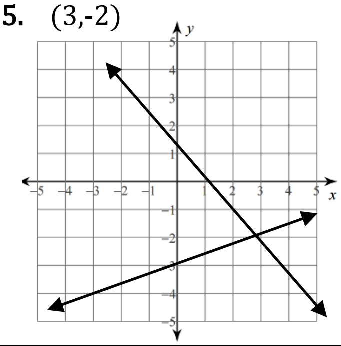
3. (0,-3)



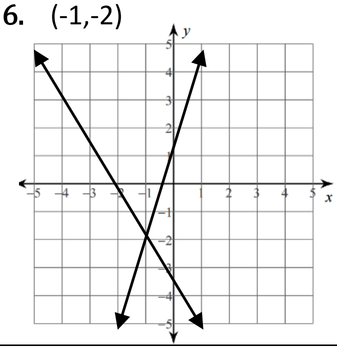
4. (-2,-2)



5. (3,-2)

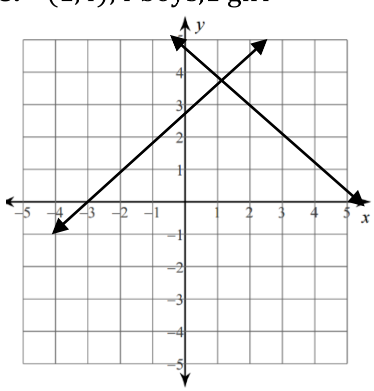


6. (-1,-2)



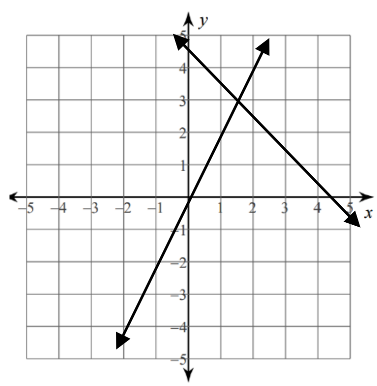
7. $x + y = 5$
$3 + x = y$

8. (1,4);4 boys,1 girl



9. $x + y = 4.50$
$y = 2x$

10. (1.5,3);notebook \$3, pencil \$1.50



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Systems of Linear Equations Practice Worksheets - ANSWER KEY

Solving Systems of Linear Equations by Substitution

1. (9,-6)	**2.** (2,6)	**3.** $(1,5)$
4. (−2,-10)	**5.** (1,4)	**6.** $\left(11,\frac{1}{4}\right)$

7. $x = 2y$ $2x + 2y = 150$ 50 ft by 25 ft	**8.** $x + y = 12$ $2x = y$ 4ft
9. $x + y = 70$ $y - x = 26$ clothes: \$48, food: \$22	**10.** $x + y = 18$ $x = y + 4$ Gertrude: 11, Elana: 7

Solving Systems of Linear Equations by Elimination

1. $(-8,6)$	**2.** $(2,\text{-}1)$	**3.** $(7,\text{-}3)$
4. $(\text{-}1,4)$	**5.** $(8,10)$	**6.** $(7,\text{-}3)$

7. $5x + 10y = 140$ $3x + 5y = 79$ \$5 each	**8.** $x + y = 100$ $2x + 3y = 240$ 60 brownies, 40 cupcakes
9. $x + y = 204$ $y = 3x$ 153 canned goods	**10.** $x + y = 6$ $25x + 5y = 110$ four 25-pound plates, two 5-pound plates

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Exceeding the CORE

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